Appendix

A Experimental Setup

For the experimental setup we combine the Natural Questions (NQ) and HotpotQA datasets for training, and validate on HotpotQA. We use Qwen2.5-3B as backbone to our model and train using group relative policy optimization (GRPO) using the verl framework. We apply GRPO with a group size of 5, a rollout temperature of 1.0, a training batch size of 512, and a validation batch size of 256. We set a maximum prompt length to 4096 tokens, a response length to 500 tokens, and an observation length of 1200 tokens. We employ a learning rate of 1×10^{-6} and a KL divergence coefficient (kl_loss_coef) of 0.001. The model is trained using 4 NVIDIA H100 GPUs.

B Step-Wise Entailment Reward

To ensure the quality of the *nli-deberta-v3-small* model for generating the next reasoning step, we performed qualitative human evaluations on random samples. Example are shown in Tables 4 and 5.

C Summary Model Performance

	End of Phase 1		**End of Phase 2**		**End of Phase 3**		**Final Evaluation**	
Regime	**Proxy**	**EM**	**Proxy**	**EM**	**Proxy**	**EM**		
Exp 1	.913	.265	.928	.195	.203	.393	.203	.393
Exp 2	.588	.285	.512	.218	.210	.241	.210	.241
Exp 3 (wp1)	.263	.188	-	-	-	-	.263	.188
Exp 4 (wp-1)	-	-	-	-	-	-	.003	.003

Table 2. Performance breakdown across training regimes, reporting the Proxy Reward and Exact Match metrics across consecutive developmental phases, including the final evaluation checkpoint.

D Prompts and Examples

Answer the given question. You must conduct reasoning inside **<think>** and **</think>** first every time you get new information. After reasoning, if you find you lack some knowledge, you can call a search engine by **<search>** *query* **</search>**, and it will return the top searched results between **<information>** and **</information>**.

If the original query is complex or involves multiple parts, you are encouraged to **decompose** it into at most 3 smaller sub-questions, separated by ##. For example: **<search>** sub-question 1 ## sub-question 2 **</search>** and it will return the top searched results between **<information>** documents sub-question 1 ## documents sub-question 2 **</information>**.

You can search as many times as you want. Only **decompose** when the question has multiple independent parts (e.g., different entities, aspects, or comparisons). Do not decompose questions that do not need it.

If you find no further external knowledge needed, you can directly provide the answer inside **<answer>** and **</answer>** without detailed illustrations. For example, **<answer>** Beijing **</answer>**.

Question: {question}

Table 3. Prompt template. The question is appended during training and inference.

Question: When does the slender man movie come out?

<think> I need to find the title of the Slender Man movie. **</think>**
<search> Slender Man movie **</search>**
<information> Doc 1(Title: "Slender Man (film)") Slender Man is a 2018 American supernatural horror film directed by Sylvain White and written by David Birke, based on the character of the same name. Doc 2(Title: "Slender Man (film)") "Slender Man" was released in the United States on August 10, 2018, by Screen Gems. Doc 3(Title: "Slender Man") In 2016, Sony Pictures subsidiary Screen Gems partnered with Mythology Entertainment to bring a Slender Man film into theatres. **</information>**
<think> I found out that "Slender Man" was released in August 2018. However, I want to double-check this information. **</think>**
<search> release date of slender man film **</search>**
<information> Doc 1(Title: "Slender Man (film)") "Slender Man" was released in the United States on August 10, 2018, by Screen Gems, grossed over $51 million worldwide. Doc 2(Title: "Slender Man (film)") It was previously slated for May 18, and then August 24, 2018. Doc 3(Title: "Slender Man (film)") "Slender Man" was released by Screen Gems on August 10, 2018. **</information>**
<think> After reviewing both sources, I confirm that the Slender Man movie was released on August 10, 2018. **</think>**
Answer: The Slender Man movie came out on August 10, 2018. My previous action is invalid. If I want to search, I should put the query between <search> and </search>. Let me try again.
<answer> August 10, 2018 **</answer>**

Table 4. Case study 1 (successful generation).

Question: Who played professor higgins in my fair lady?
<think> First, I need to recall what information I already have. Since I am trying to answer the question and do not have a list of actors yet, I will search for it. **</think>** **<search>** My Fair Lady cast of actor who plays professor Higgins **</search>** **<information>** Doc 1(Title: "My Fair Lady (film)") In London, Professor Henry Higgins (Rex Harrison), a scholar of phonetics, believes that the accent determines a person's prospects. Doc 2(Title: "Edward Mulhare") His best-known stage role was as Professor Higgins in the original Broadway production of "My Fair Lady," having taken over from Rex Harrison in 1957. Doc 3(Title: "Harry Hadden-Paton") Hadden-Paton is currently playing the lead role of Henry Higgins in the Lincoln Center Theater revival of "My Fair Lady" on Broadway. **</information>** **<answer>** I found that Stanley Holloway played the role of Professor Higgins in the Broadway production of "My Fair Lady" in 1950. That makes him the actor who played Professor Higgins in My Fair Lady. **</answer>**

Table 5. Case study 2 (unsuccessful generation).